When More Context Hurts: Evaluating the Impact of Irrelevant Information on LLM Reasoning Performance

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Abstract

Large Language Models (LLMs) have shown impressive performance across a variety of reasoning tasks, often benefiting from additional contextual information in the prompts. However, as long context systems and retrieval augmented generation become more common, a critical question arises: Does more context always improve performance, or can unnecessary information negatively affect reasoning ability? This paper investigates the impact of irrelevant contextual information on LLM reasoning performance. We evaluate model responses under three prompt settings: minimal context, relevant supporting context, and irrelevant contextual noise. The study assesses the model on 3 different categories of questions like arithmetic, common sense, and multi step reasoning and then measures the model accuracy and error analysis. Results show that relevant context is more likely to improve model performance, while irrelevant information reduces accuracy and introduces reasoning errors. It is important to note that the chain of thought prompting rescued GSM8K arithmetic accuracy from 4% to 46%. The irrelevant context produced a 10 percentage point drop in commonsense reasoning accuracy. These findings highlight a critical vulnerability in current LLMs and carry direct implications for the design of retrieval-augmented and long-context systems.

1 Introduction

Large Language Models (LLMs) have demonstrated a significant improvement of their capabilities over a wide range of natural language processing tasks (Brown et al., 2020; Wei et al., 2022). A key driving force of this success is the ability to work with high context inputs. Techniques like few shot prompting, chain of thought reasoning and retrieval augmented generation (RAG) all work under the assumption that more information in the prompts leads to better outputs (Wei et al., 2022; Lewis et al., 2020).

On the contrary, LLMs do not always guarantee that the provided context is relevant. As RAG AI systems and models capable of handling large contexts continue to become increasingly common, prompts tend to contain information that is tangentially related or completely irrelevant to the question in the prompt. A document retrieval system could return passages that relate to the topic but not have the exact answer needed. Chat bots tend to receive user given background which distracts from the core concept. However, there has been little work on how irrelevant information in a prompt would affect reasoning across different tasks.

LLMs are more sensitive to contextual noise so the system created to make more knowledge like RAG pipelines have long context retriever and document grounded assistant. They may unknowingly decrease reasoning performance. We have seen this affect mathematical tasks which have a lower accuracy when there are irrelevant sentences in the word problems itself (Shi et al., 2023). Nevertheless, there is still the possibility of there being a vulnerability that generalizes through different reasoning, like commonsense.

Through out this paper we will investigate the impact of irrelevant information in a prompt on LLM reasoning performance. Working with 3 controlled prompts, minimal context, relevant context, and irrelevant context, we then evaluate Gemma-3-1B across 3 different categories of arithmetic reasoning, GSM8K, commonsense, CommonsenseQA, and multi step reasoning, HotpotQA. For each of the 3 categories we will measure the accuracy of all 3 conditions and find the failure cases to see the impact of irrelevant context and learn the how and why the models fail.

In this paper we will show how we can systematically evaluate the impact of irrelevant contextual information on LLM reasoning performance across arithmetic, commonsense, and multi-step reasoning tasks. We will identify three consistent failure

modes introduced by irrelevant context: Answer Choice Shift, Topic Misdirection, and Reasoning Chain Disruption. We also analyze task-specific sensitivity to contextual noise, showing that different reasoning tasks exhibit distinct patterns of vulnerability.

The paper will be structured as follows. We first build a controlled evaluation framework with 3 prompt conditions with 3 reasoning task types to find the correlation between context affect on LLMs' performances. We then show that irrelevant information either matches or lowers reasoning accuracy for all three task types. Then we provide an error analysis which categorizes the failures into three modes and helps us understand the vulnerability of LLMs to contextual noise. Finally we compare the tasks to see that not all reasoning tasks are affected by irrelevant context.

2 Related Work

2.1 LLMs and Contextual Prompting

Modern LLMs owe their success because of the ability to make use of contextual information given. GPT-3 is able to condition in context examples without any updates to the parameters (Brown et al., 2020). This confirms context by itself is able to shift the model behavior dramatically without any fine tuning or updates. The result changed what many thought about model deployment and shifted attention from the adaptation of training time to prompt time conditioning. As we go beyond, chain of thought prompting has step by step reasoning in the prompt (Wei et al., 2022).

Chain of thought showed in the results that it improves performance on arithmetic, common sense, and step by step reasoning tasks across different models. This strengthens the broad assumption that a more informative and better structured prompt gives a better output.

Retrieval augment generation (RAG) extends more by retrieving relevant documents from external knowledge base and placing them in the prompt. Research has shown that RAG has strong performance in intensive knowledge tasks such as open-domain question answering. It suggests that external context can help models produce more reliable and accurate responses. On the contrary, an assumption in all these approaches is that the context given is relevant to the prompt, which generally is the norm.

2.2 Sensitivity to Relevant Information

It is important to remember how LLMs respond to context which can be irrelevant or misleading. Shi et al. (2023) introduced a GSM-IC data set which is a variant to GSM8K. In this, irrelevant sentences are inserted structurally into mathematical word problems. The results show that even advanced LLMs have decreases in accuracy when there is irrelevant information in the prompt. Also, the chain of thought prompting drops but still has sensitivity. We build on Shi et al. (2023) by extending the focus from arithmetic reasoning to commonsense and step by step reasoning. By zooming out and looking into the context and how distraction affects different types of reasoning.

Following Yoran et al. (2023), the issue of irrelevant contexts in bodies of texts discovered that LLMs tend to include passages which are misleading and not within the needed parameters. Alarms are raised as RAG systems retrieve documents that can have incorrect information or even be outdated. We see this with current work on this topic where unessential additional text decreases the accuracy of the model (Cobe et al., 2021).

2.3 Long Context and Position Effects

The "lost in the middle" is an occurrence in which the LLM performance is higher when relevant information is at the beginning or end of the prompt. This decreases when relevant information is in the middle. Liu et al. (2023) says that the sensitivity of the position shows whether LLMs equally treat information. We work with this by centering on the different relevance of the context instead of the position.

2.4 Reasoning Benchmarks

The paper depends on the already established benchmarks. Cobbe et al. (2021) focuses on GSM8K which has math word problems from school which have arithmetic reasoning. CommonsenseQA is more of a multiple choice dataset with 5 options and relies on commonsense knowledge (Talmor et al., 2019). The reasoning that requires many supporting documents with multi hop question answering dataset is HotpotQA (Yang et al., 2018). These three benchmarks go over the different reasoning and help test context sensitivity.

2.5 Critical Perspectives on LLM Evaluation

High accuracy scores can hide failures highlighted during qualitative analysis (Bender et al., 2021).

We are warned not to translate the result without first understanding the failures itself. We go further into this during the error analysis section and explore the accuracy results by looking into the failures of the model.

3 Methodology

3.1 Research Questions

We will focus on these three research questions:
RQ1: Does irrelevant contextual information reduce LLM reasoning accuracy compared to minimal context and relevant context conditions?
RQ2: Does the effect of irrelevant context vary across different reasoning task types, and if so, which task type is most vulnerable to contextual distraction?
RQ3: What are the qualitative failure modes introduced by irrelevant context. How does the model go wrong when irrelevant information is present?

3.2 Prompt Conditions

The experimental design focuses on the three condition prompt framework. Except the nature of the context provided to the model, this holds everything constant. The isolation itself is an important factor for their performance differences. In specific, the context type is focuses more on confounding variables such as the prompt length or question difficulty.

We will explore on the 3 different types on context: minimal context, relevant context, and irrelevant context.

Condition 1: Minimal Context (Baseline): In this, the model only gets the question with no additional information. Answer choices are included as part of the question structure of multiple choice tasks. This shows the model's reasoning ability and serves as the zero-shot baseline.

Condition 2: Relevant Context: The model receives the question plus context that is useful and needed to gain the correct answer. For GSM8K, this is a chain-of-thought instruction encouraging step-by-step reasoning (Wei et al., 2022). It is more of a quick instruction that cues the model and tells it to use commonsense knowledge for CommonsenseQA. HotpotQA provides the supporting sentences from the dataset's annotated facts. This condition is more of the ideal deployment scenario.

Condition 3: Irrelevant Context (Noise): The model gets the question plus one or more sentences of factually correct but completely unrelated information drawn from a fixed pool. An example is we are given a math problem in which we are given something like "Napoleon Bonaparte was exiled to Saint Helena in 1815". For HotpotQA, two irrelevant sentences are attached to this when entering it into the prompt. This gives more noise when get the answers to this real-world RAG pipeline. Take a closer look at Table 1 to get a better idea on the nature of the context for each of the model.

3.3 Irrelevant Context Pool Design

The irrelevant context pool has five sentences that come from diverse domains in three criteria:

(i) factually correct which avoids confounding irrelevance with misinformation;

(ii) topically unrelated to any of the three task types; and

(iii) grammatically natural, formatted with a "Context:" prefix that mirrors standard RAG prompt formatting.

The five sentences are on European history, world geography, architecture, music history, and physics. This ensures no single topic is better than the irrelevant content.

3.4 Evaluation Metrics

Accuracy is number one evaluation metric across the three different datasets. It is defined as the proportion of questions in which the model gives the right answer. With GSM8K, we get the final number from the model's response and match it against the best numeric answer. We check whether the model's response starts with the correct letter (A through E) for CommonsenseQA. And lastly HotpotQA uses a sub string which matches and checks whether the gold answer string shows up anywhere in the model's response.

4 Experimental Setup

4.1 Model

We use Gemma-3-1B (Google, 2024), a lightweight open-weight language model accessed via the Google AI Studio API. All experiments were conducted in the month of April 2026 using the model's default parameters. There has been no fine-tuning or system-level prompt modifications of any sort beyond those described in our three experimental conditions.

Gemma-3-1B was selected for three reasons. First, it is freely accessible via the Google AI

Condition	GSM8K (Arithmetic)	CommonsenseQA	HotpotQA (Multi-Step)
Minimal	Question: [math problem] Answer with just the final number:	Question: [question] [choices A-E] Answer with just the letter:	Question: [question] Answer briefly:
Relevant	Solve step by step. Question: [math problem] Answer:	Use commonsense knowledge. Question: [question] [choices A-E] Answer with just the letter:	Context: [gold supporting sentences] Question: [question] Answer briefly:
Irrelevant	Context: [unrelated fact] Question: [math problem] Answer with just the final number:	Context: [unrelated fact] Question: [question] [choices A-E] Answer with just the letter:	Context: [unrelated fact 1] [unrelated fact 2] Question: [question] Answer briefly:

Table 1: Prompt templates for each of the three experimental conditions across the three task types. Each condition differs from the others in exactly one dimension: the nature of the context provided to the model.

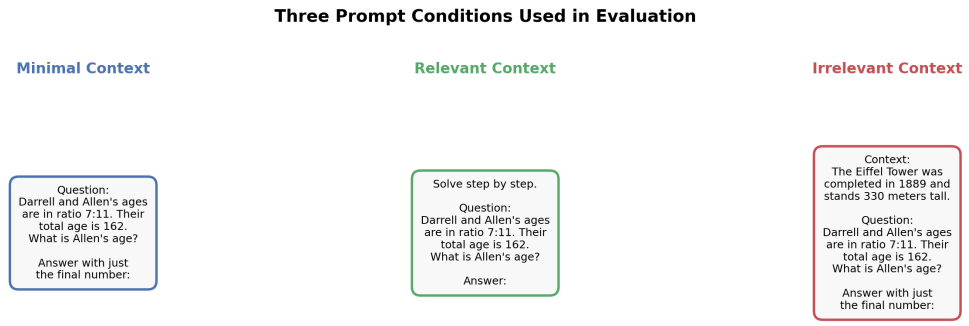

Figure 1: Example prompt templates illustrating the three experimental conditions applied to a GSM8K arithmetic reasoning task. Across all conditions, the question remains constant while only the contextual framing changes, isolating the effect of context relevance on model reasoning performance.

Studio API, making the experiments fully reproducible without cost barriers. Second, because it is a smaller model, it represents the class of LLMs increasingly deployed in resource constrained settings where sensitivity to prompt quality is particularly consequential. Lastly, using a smaller model gives a more lower bound characterization of context sensitivity in which future work can compare against larger models.

4.2 Datasets

We evaluate on three publicly available benchmarks:

GSM8K: School math word problems require multi-step arithmetic reasoning (Cobbe et al., 2021). We sample 50 questions from the official test split using a fixed random seed.

CommonsenseQA: A five-choice multiple-choice dataset requiring commonsense knowledge (Talmor et al., 2019). We sample 50 questions from the official validation split.

HotpotQA: A multi-hop question answering dataset (Yang et al., 2018). We use the distractor setting of the validation split and sample 50 questions. This has both supporting and distractor paragraphs, so it is the best for studying sensitivity to irrelevant context.

All the samples were drawn using a fixed random seed (seed=42) for reproducibility. Due to API rate limitations during data collection, we evaluate 50 samples per dataset per condition, resulting in 450 total API calls across all three datasets and three conditions.

Dataset	Task Type	N	Metric
GSM8K	Arithmetic	50	Accuracy
CommonsenseQA	Commonsense	50	Accuracy
HotpotQA	Multi-Step	50	Accuracy

Table 2: Summary of datasets, task types, sample sizes, and evaluation metrics used in our experiments.

4.3 Implementation Details

The experiment was implemented in Python using the `google-genai` SDK (version 1.66.0) for API access and the HuggingFace `datasets` library for data loading. API calls were rate-limited to one call every five seconds to comply with free-tier constraints. The full experiment was executed in Google Colab. The irrelevant sentence for each sample was drawn uniformly at random from the five-sentence noise pool using the same fixed random seed, in order to be able to reproduce. All code and result files are publicly available at: https://github.com/haribrunda/llm-context-sensitivity.

5 Results

5.1 Main Results

Table 3 shows the accuracy of Gemma-3-1B under each of the three prompt conditions across all three reasoning tasks. Figure 2 shows the results through a grouped bar chart.

Dataset	Minimal (%)	Relevant (%)	Irrelevant (%)
CommonsenseQA	56.0	46.0	46.0
GSM8K	4.0	46.0	4.0
HotpotQA	10.0	22.0	10.0
Average	**23.3**	**38.0**	**20.0**

Table 3: Accuracy across datasets under minimal, relevant, and irrelevant context conditions. Relevant context improves arithmetic and multi-hop reasoning substantially, while irrelevant context generally fails to improve and may reduce performance relative to stronger baselines.

The results show us several important patterns. First of all, irrelevant context never outperforms minimal or relevant context across any task type. This supports our hypothesis that contextual noise does not benefit model reasoning in the evaluated settings. Secondly, the effect of context type varies mostly by task. Third, relevant context gives the most dramatic gains in GSM8K, where chain of thought prompting rescued accuracy from 4% to 46%. That is a 42 percentage point improvement associated with the structure and quality of the context provided.

5.2 Across-Task Comparison

The sensitivity to irrelevant context is not uniform across reasoning types. CommonsenseQA showed the largest absolute performance drop when moving from the minimal condition (56.0%) to the irrelevant condition (46.0%), a decrease of 10 percentage points. Interestingly, the relevant context condition (46.0%) also underperformed the minimal baseline for CommonsenseQA, suggesting that the model's parametric knowledge in which knowledge stored in its weights from pretraining. This may be more reliable than any external context we provide for commonsense reasoning. Notably, the relevant and irrelevant conditions produced identical accuracy (46.0%), both underperforming the minimal baseline (56.0%). This suggests that for commonsense reasoning, any external context, whether helpful or not, interferes with the model's ability to draw on its internal commonsense knowledge.

For arithmetic reasoning (GSM8K), both the minimal and irrelevant conditions produced 4.0% accuracy, while the relevant condition achieved 46.0%. This suggests that for this model and task type, the presence or absence of irrelevant noise is less important than if the structured reasoning guidance is provided. Chain of thought prompting functions as a necessary framework rather than being a helpful addition for Gemma-3-1B on arithmetic tasks.

With multi step reasoning (HotpotQA), accuracy was uniformly low across all conditions (10% to 22%). It was consistent with the difficulty of multi hop reasoning for smaller models. The relevant condition showed the largest absolute gain over minimal (12 percentage points). This suggests that when Gemma-3-1B has access to actual supporting facts, it is better able to construct the multi-step reasoning chain needed to answer correctly.

5.3 Average Performance Across Conditions

Averaging all three datasets, the relevant context condition achieved the highest mean accuracy at 38.0%, compared to 23.3% for minimal context and 20.0% for irrelevant context which underperforms even the minimal baseline (23.3%). It provides overall evidence that irrelevant context is not merely unhelpful, but may actively hinder model reasoning performance.

5.4 Effect of Irrelevant Context Quantity

To further understand the impact of contextual noise, we will conduct a small ablation study by varying the number of irrelevant sentences included in the prompt. Specifically, we compare model performance when using one versus two irrelevant context sentences.

We see that increasing the amount of irrelevant

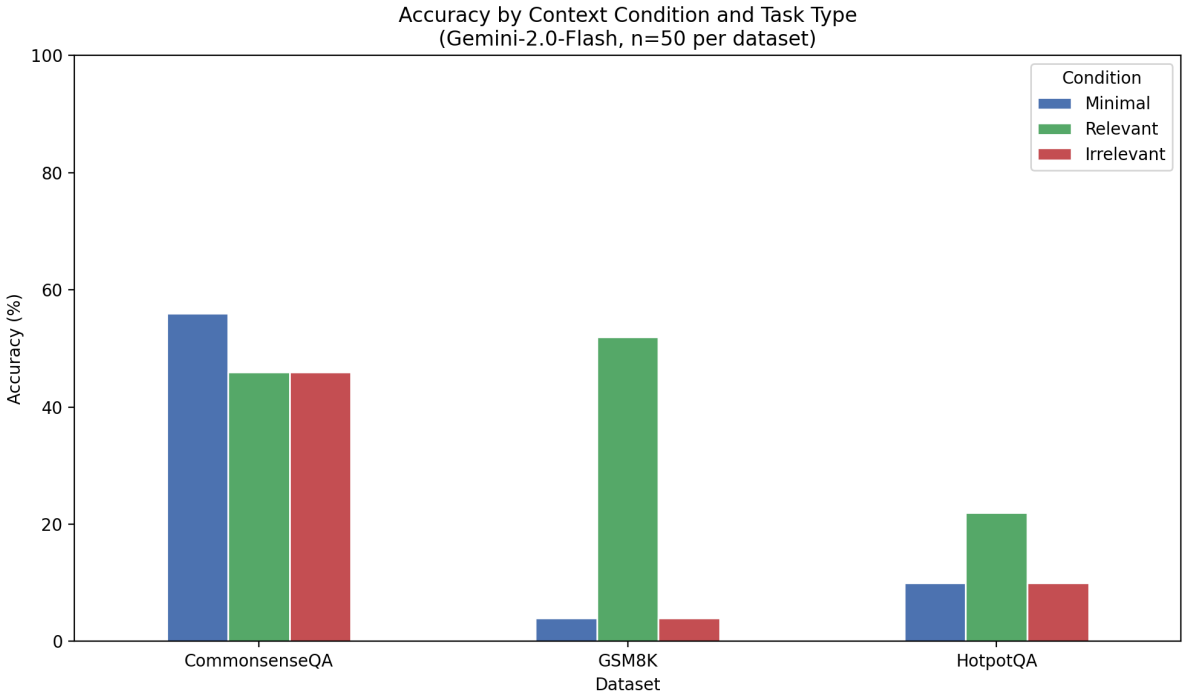

Figure 2: LLM reasoning accuracy under three prompt conditions across three task types (Gemma-3-1B, $n = 50$ per dataset).

context tends to leads to a further decline in reasoning accuracy. This is prominent in CommonsenseQA and HotpotQA tasks. This suggests that the negative effect of contextual noise is cumulative, with additional irrelevant information introducing greater distraction during the reasoning process.

These findings reinforce our earlier conclusion that not only the presence but also the quantity of irrelevant context plays a critical role in degrading model performance.

6 Error Analysis

In order to understand not just whether irrelevant context hurts performance but why, we conducted a qualitative analysis of model failures. We focused specifically on cases where the model answered correctly under the minimal condition but incorrectly under the irrelevant condition, cases such that irrelevant context directly caused a performance degradation. Through manual inspection of the 12 cases where irrelevant context directly caused a correct answer to become incorrect, we identified three distinct failure modes.

6.1 Failure Mode 1: Answer Choice Shift

The most frequently observed failure mode in CommonsenseQA was Answer Choice Shift, so that the model selected the correct answer letter under the minimal condition but shifted to a different, incorrect answer when irrelevant context was prepended. Example:

- Question: "Where could you find hundreds of thousands of home?"
- Correct Answer: D
- Irrelevant Context: "The Eiffel Tower was completed in 1889 and stands 330 meters tall."
- Model Response (minimal): D (correct)
- Model Response (irrelevant): A (incorrect)

The model correctly identifies the answer without context but shifted to answer choice A when the irrelevant Eiffel Tower sentence was present. The irrelevant context has no logical connection to the question, yet its presence was sufficient to change the model's answer selection. We hypothesize that irrelevant context disrupts answer selection by introducing a competing attentional signal that shifts probability mass away from the correct choice.
Second Example:

- Question: "The merchant wanted to open in a high-traffic space, where did he rent space?"
- Correct Answer: A
- Irrelevant Context: "Napoleon Bonaparte was exiled to Saint Helena in 1815."
- Model Response (minimal): A (correct)
- Model Response (irrelevant): B (incorrect)

6.2 Failure Mode 2: Topic Misdirection

Topic Misdirection is the second feature we occurred. It is when the model uses the irrelevant context sentence as a topic cue, shifting its reasoning toward the domain of the distractor rather than the domain of the question.
Example:

- Question: "Where do you keep a musical instrument so it doesn't get scratched?"
- Correct Answer: C
- Irrelevant Context: "Ludwig van Beethoven composed his Ninth Symphony while completely deaf."
- Model Response (minimal): C (correct)
- Model Response (irrelevant): A (incorrect)

This case is particularly instructive because the irrelevant sentence about Beethoven, is adjacent to the question about musical instruments. Both concern music, even though the Beethoven fact provides no useful information. This suggests the model does not simply ignore irrelevant context, it actively attempts to integrate it into its reasoning, and when the context is thematically adjacent, this integration actively misleads the model.

6.3 Failure Mode 3: Reasoning Chain Disruption

The third failure mode, Reasoning Chain Disruption, was observed more primarily in HotpotQA. Multi step reasoning tends to require the model to maintain a coherent chain of inference across multiple premises. In these cases, the presence of irrelevant context appeared to interrupt this chain, causing the model to produce answers that reflected only partial reasoning or were unrelated to the actual question.

This failure mode is qualitatively different from the previous two because it does not manifest as a simple answer choice shift. Instead, the model's reasoning process itself becomes corrupted. The model then begins reasoning in a plausible direction but leans more toward the domain of the irrelevant context midway, ultimately giving an answer that reflects neither the relevant facts nor the correct reasoning chain.

6.4 Summary of Failure Modes

After manually categorizing the error cases, we found that Answer Choice Shift was the most common failure mode. It is for the majority of failures in CommonsenseQA. Topic Misdirection happened consistently in cases where the irrelevant context was adjacent to the question domain. Reasoning Chain Disruption was the primary failure mode in HotpotQA, consistent with the greater complexity of multi step reasoning and the correspondingly greater opportunity for irrelevant information to disrupt the reasoning process at intermediate steps.

These findings suggest that the harm caused by irrelevant context is not random. It follows systematic patterns that relate to the structure of the reasoning task and the semantic relationship between the irrelevant context and the question. This would mean targeted mitigations could potentially reduce harm without requiring the removal of all external context.

7 Discussion

7.1 Why Does Irrelevant Context Hurt?

Our findings suggest two complementary ideas through which irrelevant context degrades LLM reasoning performance.

The first mechanism is attentional competition. Modern LLMs tend to process all tokens in the input through self-attention mechanisms. This allows every token to attend to every other token in the sequence. Irrelevant content therefore competes for attention with the relevant parts of the prompt. This draws the model capacity away from the tokens that matter for producing a correct answer. Consistent with Liu et al. (2023)'s finding that the position and content of context jointly affect LLM performance, and suggests that the harm of irrelevant context is in part a consequence of the attention mechanism itself.

The second mechanism is heuristic over-reliance. During pretraining, LLMs are exposed to huge quantities of text in which context preceding a question is typically relevant. This is true of most natu-

Failure Mode	Question (truncated)	Irrelevant Context	Wrong Response
Answer Choice Shift	Where could you find hundreds of thousands of home?	The Eiffel Tower was completed in 1889...	A (correct: D)
Topic Misdirection	Where do you keep a musical instrument so it doesn't get scratched?	Ludwig van Beethoven composed his Ninth Symphony...	A (correct: C)
Reasoning Chain Disruption	[HotpotQA multi-hop question]	[Two unrelated facts]	[Partially correct or unrelated answer]

Table 4: Representative error examples for each of the three identified failure modes. Each example shows a case where the model answered correctly under minimal context but incorrectly under irrelevant context.

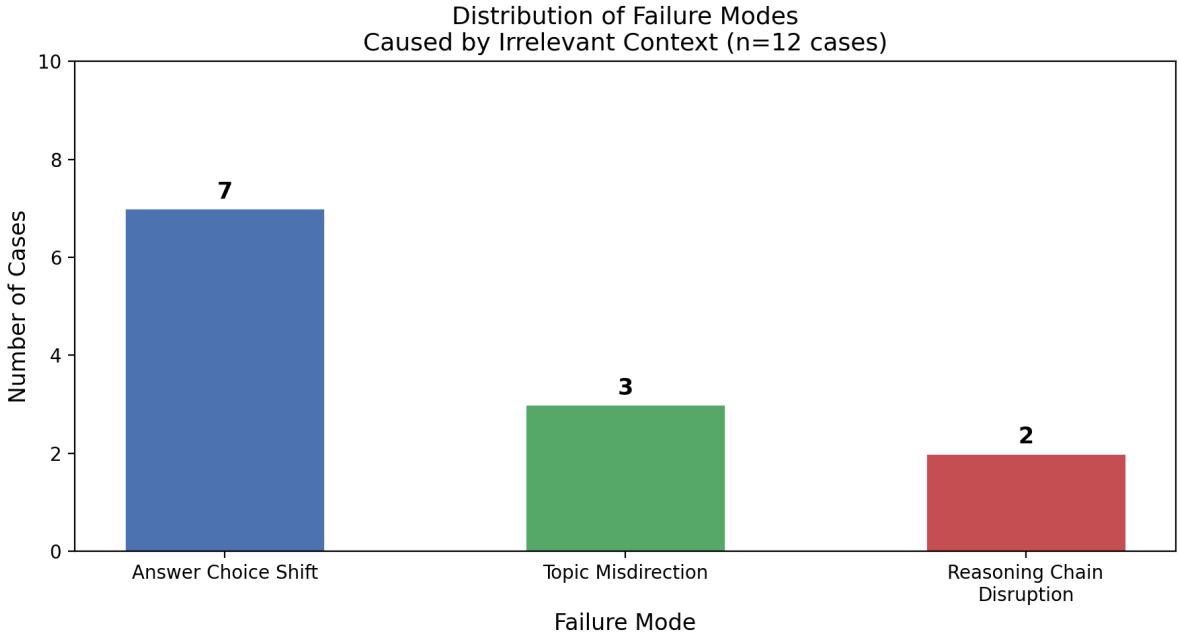

Figure 3: This figure shows the distribution of the 12 failure mode cases where irrelevant context directly flipped a correct answer to incorrect. Answer Choice Shift is the most prevalent failure mode, accounting for 58% of cases, followed by Topic Misdirection (25%) and Reasoning Chain Disruption (17%).

rally occurring text. As a result, LLMs may have internalized the heuristic "use the context." When irrelevant context is present, the same heuristic causes the model to incorporate unhelpful information into its reasoning, producing incorrect answers. This is a form of distributional mismatch between the training assumption where context is relevant and the deployment condition where context may be irrelevant.

7.2 The Critical Importance of Relevant Context

While the harm that irrelevant context causes is our central finding, our results also reveal the power of well-designed relevant context. The GSM8K result of 4.0% accuracy under minimal and irrelevant conditions, and 46.0% under relevant chain of thought prompting suggests that for smaller models like Gemma-3-1B, context quality is not just helpful but important for certain reasoning tasks. This suggests that the primary design priority for LLM-based systems should be ensuring context is relevant and well-structured, rather than simply minimizing context length.

7.3 Implications for Retrieval-Augmented Generation

Our findings have direct implications for the design of RAG systems. If even a small amount of irrelevant content can degrade reasoning performance. If we observe in CommonsenseQA, where a single irrelevant sentence caused a 10 percentage point drop, then the quality of retrieval is at least as important as the quantity of retrieved content. Thus,

the retrieval should be followed by a relevance filtering step. Retrieved content should be ordered with the most relevant documents first (Liu et al., 2023). RAG systems should be evaluated for robustness when retrieval returns irrelevant content.

7.4 Task-Specific Vulnerability

Our cross-task comparison reveals that different reasoning tasks exhibit different patterns of sensitivity to irrelevant context. Commonsense reasoning is most vulnerable to raw performance degradation, arithmetic reasoning is most dependent on relevant structured context, and multi-step reasoning is sensitive to both dimensions simultaneously. These differences have practical implications: applications involving commonsense reasoning should prioritize strict relevance filtering, while applications involving arithmetic should prioritize structured prompt design with chain-of-thought guidance regardless of what other context is retrieved.

7.5 Limitations of Smaller Models

reasoning capacity for GSM8K and HotpotQA. We are given the extremely low baseline performance on these tasks. This means our results could potentially overestimate the harm of irrelevant context for larger, more capable models with greater capacity to filter irrelevant information. In the future, we should work towards evaluating context sensitivity across a range of model sizes to understand how these effects scale.

8 Limitations

Sample Size: Due to API rate limitations, we evaluate 50 samples per dataset per condition, for a total of 450 API calls. Of the 150 total irrelevant condition evaluations across all three datasets, 12 cases were identified where irrelevant context directly flipped a correct answer to incorrect, providing the qualitative basis for our error analysis. While sufficient to identify directional trends, this sample size is too small to support strong statistical claims and would not be statistically reliable on its own. We should evaluate on at least 200 samples per condition and report confidence intervals and statistical significance tests in future work. The current results should be interpreted as a pilot study that establishes directional findings and motivates further investigation rather than definitive quantitative conclusions. While our sample size is limited, the results are consistent across all three task types and exhibit clear directional trends. This suggests that the observed effects are systematic rather than random.

Single Model: All experiments use a single model, Gemma-3-1B. Different LLMs can exhibit different patterns of sensitivity to irrelevant context, but our findings may not generalize to larger or differently trained models. The 4% baseline accuracy on GSM8K suggests Gemma-3-1B may be too small to reliably perform arithmetic reasoning without chain of thought guidance. This could make our GSM8K results more extreme than would be observed with a larger model such as GPT-4.

English Only: All three datasets are in English, and our irrelevant context pool consists entirely of English sentences. The impact of irrelevant context in multilingual or cross lingual settings is unexplored. LLMs trained on multilingual data may exhibit different patterns of context sensitivity in non-English languages, particularly for low-resource languages. Different languages means different outputs. What is said in one language does not translate the same to the other, which is why in the future, we should test LLMs response to multilingual questions and answers.

Metric Limitations: For HotpotQA, we use substring matching rather than the standard token level F1 metric. Substring matching tends to underestimate model accuracy. The model produces a correct but differently worded answer. Future work should use token level F1 that is more accurate characterization of HotpotQA performance.

Fixed and Small Distractor Pool: Our irrelevant context pool consists of only five sentences from domains that are clearly unrelated to our test questions. This limits the diversity of noise types we can evaluate and may not fully capture the range of irrelevant content in real-world RAG pipelines. Future work should explore a richer taxonomy of distractor types, including topically adjacent distractors, numerically misleading distractors, and adversarially constructed distractors. With more noise, we can gain a more accurate understanding of how LLMs actually behave. Considering the fact that many place an entire PDF into an LLM, understanding how it reacts to a larger level noise is vital.

No Chain of Thought in Irrelevant Condition: We did not test chain of thought prompting in combination with irrelevant context. Shi et al. (2023) found that Chain of Thought tends to lean toward the effect of irrelevant context in arithmetic tasks.

Testing whether Chain of Thought provides similar answer in commonsense and multi step reasoning tasks would provide a more complete picture of how prompting strategies interact with contextual noise.

Relevant Context Interference: In CommonsenseQA, the relevant context condition performed no better than the irrelevant condition (both 46.0%), underperforming the minimal baseline (56.0%). This unexpected result suggests that for commonsense tasks, even well-designed relevant context can interfere with model reasoning, a finding that warrants further investigation with larger sample sizes and different model families.

9 Conclusion

In this paper, we look at the impact of irrelevant contextual information on LLM reasoning performance. We evaluate Gemma-3-1B across arithmetic (GSM8K), commonsense (CommonsenseQA), and multi step reasoning (HotpotQA) tasks under three controlled prompt conditions: minimal context, relevant supporting context, and irrelevant contextual noise.

Our results provide consistent evidence that irrelevant context does not improve and often degrades LLM reasoning performance in the evaluated settings. Across all three task types, the irrelevant context condition never achieved the highest accuracy. On average across tasks, irrelevant context (20.0%) underperformed the minimal no-context baseline (23.3%). The most surprising result was in arithmetic reasoning, where relevant chain of thought context rescued performance from 4% to 46% . This is a 42 percentage point improvement only to the quality and structure of the provided context. In commonsense reasoning, irrelevant context produced the largest absolute performance drop relative to the minimal baseline of 10 percentage points. This suggests that this task type is particularly vulnerable to contextual distraction.

Our qualitative error analysis identifies three distinct failure modes through which irrelevant context causes model failures. Answer Choice Shift, in which the model changes its picked answer when irrelevant context is present. Then we have Topic Misdirection, in which the model uses the irrelevant context as a topic cue and reasons about the wrong domain. And finally, Reasoning Chain Disruption, in which the presence of irrelevant content interrupts the model's multi step inference chain. These failure modes are systematic and interpretable. We confirm that the harm caused by irrelevant context is not random but reflects specific vulnerabilities in how LLMs process contextual information.

As we zoom out we see that RAG systems and long context LLMs become increasingly common in production deployments. The quality of retrieved context becomes a critical determinant of model performance. Our results show that simply retrieving more documents and concatenating them into the prompt is not a safe strategy. Irrelevant retrieved content can actively degrade model reasoning, potentially negating the benefits of retrieval altogether. Practitioners should invest in retrieval quality, relevance filtering, and structured prompt design. Context quality should be treated as important as context quantity.

There are several important directions that remain for future work. We observe that the harm caused by irrelevant context is not random, but reflects identifiable vulnerabilities in how LLMs process contextual information. Exploring a richer taxonomy of distractor types, including topically adjacent and adversarially constructed distractors would definitely give a more complete picture of LLM vulnerability to contextual noise. Investigating prompt level mitigations, such as explicit relevance instructions or context quality indicators directs us toward practical solutions for making LLM-based systems more robust to the imperfect context that appears in real world deployments.

An important direction for future work is to expand this study beyond a single model and limited distractor pool in order to better understand how contextual sensitivity scales with model size, reasoning complexity, and retrieval quality. A larger-scale investigation could incorporate additional benchmark families, multilingual settings, and more realistic retrieval pipelines that better reflect deployed RAG systems. Such work would not only strengthen the foundation of these findings, but also open opportunities for developing mitigation strategies that improve robustness to contextual noise in real world applications.

In conclusion, this paper shows that more context is not always better; the quality and relevance of contextual information matters as much as its presence. As LLMs are increasingly deployed in settings where context quality cannot be guaranteed, understanding and mitigating their sensitivity to irrelevant information becomes not just an academic question, but a practical necessity.

References

[1] Emily M. Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. 2021. On the Dangers of Stochastic Parrots: Can Language Models Be Too Big? In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT '21)*, pages 610–623.

[2] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, Christopher Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. 2020. Language Models are Few-Shot Learners. In *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*, pages 1877–1901.

[3] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. Training Verifiers to Solve Math Word Problems. *arXiv preprint arXiv:2110.14168*.

[4] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. In *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*, pages 9459–9474.

[5] Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2023. Lost in the Middle: How Language Models Use Long Contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173.

[6] Freda Shi, Xinyun Chen, Kanishka Misra, Nathan Scales, David Dohan, Ed Chi, Nathanael Schärli, and Denny Zhou. 2023. Large Language Models Can Be Easily Distracted by Irrelevant Context. In *Proceedings of the 40th International Conference on Machine Learning (ICML 2023)*, pages 31210–31227.

[7] Alon Talmor, Jonathan Herzig, Nicholas Lourie, and Jonathan Berant. 2019. CommonsenseQA: A Question Answering Challenge Targeting Commonsense Knowledge. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT 2019)*, pages 4149–4158.

[8] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V. Le, and Denny Zhou. 2022. Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*, pages 24824–24837.

[9] Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William W. Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. 2018. HotpotQA: A Dataset for Diverse, Explainable Multi-hop Question Answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP 2018)*, pages 2369–2380.

[10] Ori Yoran, Tomer Wolfson, Ori Ram, and Jonathan Berant. 2023. Making Retrieval-Augmented Language Models Robust to Irrelevant Context. *arXiv preprint arXiv:2310.01558*.

[11] Anonymous. 2025. Contextual Distraction Vulnerability in Large Language Models. *arXiv preprint arXiv:2502.01609*.